

Characterization of bio-oil and its sub-fractions from pyrolysis of *Scenedesmus dimorphus*



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ABSTRACT

In the present study, microalgae *Scenedesmus dimorphus* was reported for pyrolysis in a fixed-bed reactor to determine the effects of temperature on products yield and the chemical compositions of the liquid and solid products. Experiments were carried out at a temperature range of 300–600 °C with heating rate of 40 °C/min and nitrogen flow rate of 100 ml/min. The yield of bio-oil was found to be maximum (39.6%) at the temperature of 500 °C and was further fractionated into *n*-hexane, toluene, ethyl acetate and methanol sub-fractions by using liquid column chromatography. Various characteristics of bio-oil and its sub-fractions were determined by ¹H NMR, FTIR and GC–MS. The biochar produced as a co-product can be a potential soil amendment with multiple benefits including soil fertility and C-sequestration. The present investigation suggests the suitability of *Scenedesmus dimorphus* as a potential feedstock for exploitation of energy and biomaterials through pyrolytic conversion.

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1. Introduction

Fossil fuel shortage and severe environmental problems have attracted great attention towards exploration of clean renewable energy. Biomass as a renewable energy source appears to be one of the most effective solutions for clean and sustainable energy, so as to overcome the problem of global warming and reduce the fossil fuel consumption. Moreover, it also has additional advantage of low sulfur and nitrogen content [1]. Among the various biological resources, microalgae are considered as one of the most promising feedstocks for replacing conventional petroleum-based energy fuels. High biomass productivity, non-competition with food crops, ability to capture carbon dioxide and by-product from wastewater remediation are some of the advantages of microalgae [2]. Theoretically, microalgae can produce 100 times higher oil per hectare than that of terrestrial crops [3,4]. Microalgae consume 2 g of CO₂ to produce 1 g of biomass during conversion of glucose into starch or lipids. Thus, by assuming a growth rate of 50 g/m²/day, it is possible for 1 ha area of pond containing algae to absorb up to one ton of CO₂ per day [5].

Biomass resources can be converted into solid, liquid and

gaseous fuel by applying suitable conversion technologies, such as biochemical and thermochemical conversion routes. Due to its ability to utilize almost all types of biomass and recover both energy and chemical value of the feedstock; thermochemical conversion technologies have the advantage over biochemical conversion technologies. In this regard, pyrolysis is of great interest since the product selectivity can be controlled and tuned to suite end-use interests by regulating the various process parameters [6]. Along with the liquid and gaseous products, a solid carbonaceous product is also co-produced during pyrolysis (biochar). Biochar can be used in a variety of ways: as a soil amendment, energy carrier, adsorbents, catalyst support. Use of biochar as a soil amendment addresses two global issues viz., climate change and sustainable soil management [7]. According to Lehmann [8], biochar application in soil improves soil fertility, enhances water holding capacity, increases soil pH and nutrient retention capacity of soil. Biochar helps in mitigation of climate change through long term carbon sequestration which otherwise would have been released in the atmosphere due to decomposition of biomass. In addition to all this, it is a safer method of organic waste management [9].

Conversion of various microalgal species like, *Oscillatoria tenuis* [10], *Chlorella protothecoides* [10,11], *Spirulina platensis* [12], *Chlorella protothecoides*, *Microcystis aeruginosa* [13], *Chlorella vulgaris* and *Synechococcus* [14], *Dunaliella teriolecta* [15] to biofuel by pyrolysis have been reported. Miao et al. [16] reported the fast

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pyrolysis of microalgae *Chlorella protothecoides* and *Microcystis aeruginosa* in a fluid bed reactor where microalgae was feed at a rate of 4 g/min. The bio-oil obtained had higher heating value of 29 MJ/kg, which was about 1.4 times more than wood. The saturated and polar fractions accounted for 1.14 and 31.17% of the bio-oils from microalgae on an average, which were higher than the wood derived bio-oil. Chaiwong et al. [17] reported that bio-oil and biochar produced from *Spirulina Sp* had maximum degradation temperature at 322 °C. The energy consumption ratio (ECR) of bio-oil and biochar was 0.49 and the net energy output was positive. Miao and Wu [18] reported an approach for increasing the bio-oil yield from fast pyrolysis by manipulating the metabolic pathway in microalgae through heterotrophic growth. The yield of bio-oil (57.9%) from heterotrophic *Chlorella protothecoides* cells was 3.4 times higher than autotrophic cells by fast pyrolysis. The bio-oil had much lower oxygen content, with a higher heating value (41 MJ/kg), lower density (0.92 kg/l) and lower viscosity (0.02 Pa) compared to the bio-oil from autotrophic cells and wood. Ferreira et al. [19] reported the production of bio-oil from microalgae (*Chlorella vulgaris* and *Scenedesmus obliquus*) through pyrolysis in a fixed bed reactor with and without catalyst. Both the microalgae showed significant differences in bio-oil yields in the range of 26–38 wt% and 28–50 wt% respectively. The use of carbonate catalysts increased the gas yield and decreased the bio-oil yield for both microalgae. The use of alkaline salt catalysts improved the quality of the obtained bio-oil with a net gain on the alkyl groups. Phukan et al. [20] reported that microalgae *Chlorella sp.* MP-1 exhibited several important attributes for futuristic study on renewable energy. With simple and inexpensive nutrient regime to culture, faster growth rate as compared to terrestrial energy crops, high biomass productivity, attractive biochemical profile and good energy content (18.59 MJ/kg), *Chlorella sp.* MP-1 offered a strong candidature.

The North-eastern region of India, particularly Assam, is known for having a rich biodiversity of freshwater microalgae. Many microalgal species have been screened and reported for oil content in this region. Among the microalgae species *Chlorella sp.*, *Selenastrum sp.*, *Scenedesmus dimorphus*, *Scenedesmus quadricauda*, *Desmodesmus sp.* have good potential for high biomass production and to develop technology for mitigation of harmful green house gases [21]. Thus, there remains immense scope for establishing decentralized pyrolysis based bio-refineries in rural north-east India with *Scenedesmus dimorphus* and other abundantly available bio-resources as feedstock due to the fact that pyrolysis based bio-refineries are comparatively easier to build and operate.

Among microalgae, *Scenedesmus* genus has the most desirable features for efficient and economic combination of CO₂ fixation, wastewater treatment and lipid synthesis. Okkou et al. [22] optimized the best culture conditions for the highest biomass production for *S. dimorphus*. Nitrogen was quantitatively the most important nutrient for production of biomass and lipid. Recently, *S. dimorphus* drew attention as a commercially viable source for wide spectrum of compounds such as high quality protein, B₁₂, C and E vitamins, pigments and other bioactive chemicals, long chain polyunsaturated fatty acids especially “omega-3” and “omega-6”, glycolipids, sulfolipids and phenolic compounds. *S. dimorphus* seems to be the ideal species, due to its ability to grow and tolerate different environmental conditions, and is more resistant to physiological stress consequent to nutrient starvation and light limitation [23].

In the present study, characterization of bio-oil and biochar obtained from pyrolysis of microalgae *Scenedesmus dimorphus* was carried out. Pyrolysis of *S. dimorphus* was performed at temperatures ranging from 300 to 600 °C with a heating rate of 40 °C/min. The effect of pyrolysis temperature on product yield was also

studied. The liquid products were analyzed for different physical and chemical properties by using different chromatographic and spectroscopic technique such as FT-IR, ¹H NMR, GC/MS and column chromatography. The co-product of pyrolysis i.e. biochar obtained at different temperatures were also analyzed for different physical and chemical properties.

2. Experimental section

2.1. Raw material and its characterization

The raw *S. dimorphus* microalgae contains 76% water. Dried *S. dimorphus* sample was obtained from Defense Research Laboratory, Defence Research & Development Organization, Tezpur, India. The sample was crushed in a pulverizer and allowed to pass through 0.2 mm (70 mesh) (as per TAPPI T257 Om-85 methods). The ground sample was then oven dried at 60 °C for over-night and then kept for cooling in desiccators for 8 h to get the desired dried biomass.

Proximate and ultimate analyses were carried out for the microalgae sample *S. dimorphus*. Proximate analysis gives the information on moisture content, ash content, volatile matter content and fixed carbon content of the material and was conducted following the methodology of ASTM (American Standard Test Method D3173 and D3175). For moisture content, samples were dried at 105 ± 3 °C until a constant weight was reached (i.e. oven dry); for volatile matter samples were heated to 950 °C for 2 min (covered crucible) and for ash content samples were heated in muffle furnace at 575 ± 25 °C for a period of 3 h (uncovered crucible). Ultimate analysis was performed to determine the elemental composition of the material. It was carried out by using an elemental analyzer (Euro EA elemental analyzer) which provides carbon, hydrogen, nitrogen, sulfur percentage. The oxygen content was calculated by difference. The microalgae feedstock was analyzed for total carbohydrates by using Anthrone method according to Pons et al. [24], total protein according to Lowry et al. [25] and total lipid content according to Bligh and Dyer [26] method. The calorific value of sample was determined using a bomb calorimeter (5E-1AC/ML, Auto bomb calorimeter) according to ASTM D2015. Thermo-gravimetric analysis (TGA) of the sample was done using a DTG60 instrument. About 5 mg sample was placed in a standard alumina crucible. The TG experiments were performed under flowing nitrogen (N₂) atmosphere with a flow rate of 100 ml/min. The temperature was increased from 50 to 900 °C with a heating rate of 10 °C/min. Thermo-gravimetric weight loss curve was plotted against temperature.

2.2. Experimental procedure

The pyrolysis experiments were carried out in a fixed bed vertical, tubular reactor (length: 30 cm and internal diameter: 2.47 cm) made of quartz glass in which the temperature was controlled by Ni–Cr thermocouple placed in the center of electrical furnace as reported elsewhere [6]. For each pyrolysis experiment, 10 g of the biomass sample was taken in the lab scale reactor at a bed height of 4 cm. The experiments were carried out with a temperature increment of 40 °C/min to final temperature of 300, 400, 500, 600 °C, under constant flow rate of nitrogen at 100 ml/min. A condenser was attached at the outlet of the reactor to condense the vapors coming out of it where water was used as cooling medium. The condensed vapors were collected in a container as liquid product whereas there was some amount of non-condensable gases which were simply left out. The yields of liquid products and char were determined by weighing. The amount of gas was determined by difference. The liquid product consists of two

phases: aqueous phase and organic phase. The aqueous phase was separated from the oil phase in a separating funnel using solvent, diethyl ether. The oil phase was then dried over anhydrous sodium sulfate, filtered and evaporated in a rotary evaporator at 30 °C. Upon removal of diethyl ether, this fraction was weighted and termed as bio-oil which was analyzed using various analytical techniques.

The following equations were used to calculate the product yields of pyrolysis experiments:

$$\text{Oil yield (wt\%)} = \frac{W_{\text{Oil}}}{W_{\text{Biomass,db}}} \times 100$$

$$\text{Aqueous yield (wt\%)} = \frac{W_{\text{Aqueous}}}{W_{\text{Biomass,db}}} \times 100$$

$$\text{Solid yield (wt\%)} = \frac{W_{\text{Solid,db}}}{W_{\text{Biomass,db}}} \times 100$$

$$\text{Gas yield (wt\%)} = 100\% - \text{Liquid yield (wt\%)} - \text{Solid yield (wt\%)}$$

where, W = the weight and db = dry basis.

2.3. Fractionation of bio-oil

The chemical classes of compositions of the bio-oils were determined by fractionation using liquid column chromatography

$$\text{CS(\%)} = \frac{\text{M(g)} \times \text{char yield \%} \times \text{carbon\% biochar} \times \text{R(50) index of biochar}}{\text{M(g)} \times \text{carbon\% of feedstock}}$$

as reported elsewhere [27]. The column (70 cm × 1.5 cm) was packed with activated silica gel (70–230 mesh) up to 3/4 of the total volume and pretreated at 105 °C for 2 h prior to use. First the column was eluted with 500 ml *n*-hexane to obtain the *n*-hexane soluble fraction. After that the same column was eluted with toluene and ethyl acetate and at the end the column was washed with methanol. Collecting bottle was changed when the apparent color ribbon layer reached the end of the column. Rotavapor was run by maintaining the temperature at 40 °C and reduced pressure at 28 mmHg. The solvents remained after rotary evaporation was dried by using freeze dryer and then weights of sub-fractions were calculated and subjected to analysis for the present study [28].

2.4. Characterization of biochar

Proximate and ultimate analyses of the biochar were carried out according to ASTM D3172-07a. The calorific value of biochar was determined using a bomb calorimeter (5E-1AC/ML, Auto bomb calorimeter) according to ASTM D2015. Scanning electron microscopy (SEM) images of the char obtained at 500 °C was taken with a JEOL (JSM-6390 LV) microscope with an acceleration voltage of 20 kV. SEM analysis was performed to find out the surface morphology of the biochars. Functional groups were determined by using Nicolet Impact I-410 model IR spectrometer. pH values were measured by adding biochar to de-ionized water in a mass ratio of 1:20. The solution was then hand shaken and allowed to stand for 5 min before measuring the pH with a pH meter (EUTECH Instruments pH 700). Electrical conductivity (EC) were measured with conductivity meter (Digital TDS/Conductivity Meter MK509)

using the same solution as prepared for pH measurement. The purpose for measurement of pH was to understand how application of biochar can influence soil acidity. This is particularly important as a soil of north-east India is generally acidic in nature. EC values of biochar is significant as it is directly proportional to the amount of salt (ion) concentration present in biochar and is therefore, related to the buffering capacity and cation exchange capacity (CEC) of biochar. Surface area of the biochar was measured by using AS-3012 BET Surface Area Analyzer. Carbon sequestration of the biochar was also determined by considering the R₅₀ index. Carbon sequestration potential (CSP) can be considered as the final amount of carbon that is captured in the soil. CSP was calculated by subtracting carbon loss during pyrolysis from C present in the raw biomass and multiplying it by recalcitrance (R₅₀), where weight of the feedstock can be taken as M (g). R₅₀ index was used to evaluate the thermal recalcitrance of biochar and was obtained from TGA analysis as proposed by Harvey et al. [29].

$$R_{50, \text{ of biochar}} = \frac{T(50)\text{biochar}}{T(50)\text{graphite}}$$

where T_{50, biochar} and T_{50, graphite} are the temperature at which 50% weight was lost by oxidation/volatilization of biochar and graphite, respectively in atmosphere, values were obtained from TG thermograph [30].

2.5. Characterization of liquid product

Physical properties such as density, specific gravity, flash point, fire point, pour point, cloud point and calorific value of bio-oil were determined by using ASTM methods as reported elsewhere [31,32]. The pH of the pyrolytic oil was determined by using pH meter (EUTECH pH 510). Bio-oil was characterized by using Euro EA elemental analyzer to determine bio-oil's elemental contents C, H, N and S. Fourier transforms infrared spectroscopic (FTIR) analysis of the bio-oil and its sub-fractions were recorded on a Nicolet Impact I-410 model IR spectrometer to determine its functional groups. The ¹H NMR spectra of the bio-oil and its sub-fractions were recorded in a 400 MHz NMR spectrophotometer (JEOL, JNM ECS) by using deuterated chloroform (CDCl₃) as the internal standard. The bio-oil samples were analyzed by GC–MS (Perkin Elmer Claurus 600) equipped with TCD detector and DB-5ms column (60.0 × 250 μm). Helium (99.99%) was used as the carrier gas with constant flow of 1.0 ml/min. The oven temperature was programmed from 70 °C (2 min) to 300 °C at a heating rate of 10 °C/min with a hold time of 10 min. A sample volume of 3 μL was injected. The MS was operated in electron ionization mode and an m/z range from 24 to 624 was scanned.

3. Results and discussion

3.1. Characterization of microalgae sample (*S. dimorphus*)

3.1.1. Proximate and ultimate analysis of microalgae

The results of chemical composition, proximate and ultimate

Table 1
Properties of *S. dimorphus* microalgae and bio-chars obtained at different temperatures.

Properties	Raw biomass	Biochar			
		300 °C	400 °C	500 °C	600 °C
pH	—	7.18 ± 0.07	7.93 ± 0.17	7.96 ± 0.22	8.32 ± 0.02
EC (m mho cm ⁻¹)	—	0.14 ± 0.03	0.15 ± 0.03	0.24 ± 0.03	0.46 ± 0.04
Protein content (wt.%)	53.13	—	—	—	—
Lipid content (wt.%)	10.23	—	—	—	—
Carbohydrate content (wt.%)	22.90	—	—	—	—
Proximate analysis (dry basis, wt.%)					
Moisture Content	0.083	0.08 ± 0.003	0.07 ± 0.007	0.06 ± 0.004	0.05 ± 0.005
Ash Content	17.33	39.60 ± 0.26	41.43 ± 0.95	43.30 ± 0.36	44.17 ± 0.42
Volatile Content	51.45	17 ± 0.26	6.63 ± 0.29	4.33 ± 0.60	3.33 ± 0.25
Fixed C (by difference)	31.13	43.40 ± 0.82	51.87 ± 0.64	52.37 ± 0.70	52.17 ± 0.68
Ultimate analysis (drybasis, wt.%)					
C	52.60	46.82	47.37	50.20	53.58
H	6.21	7.85	7.93	8.01	8.50
N	8.75	5.85	5.63	6.01	6.50
O (by difference)	31.92	39.48	39.07	35.78	31.42
H/C	1.42	2.01	2.01	1.91	1.90
O/C	0.46	0.63	0.62	0.53	0.44
HHV (MJ/Kg)	19.0	—	—	—	—
R ₅₀ index	—	0.62	0.67	0.70	0.76
CSP%	—	25.26	32.98	34.49	36.27

analysis of *S. dimorphus* sample were furnished in Table 1. The protein and carbohydrate contents of the microalgae biomass were 53.13% and 22.90% respectively and were in agreement with results reported by Kumar et al. [33]. The proximate analysis gave the moisture content, volatile matter (VM), ash content of the sample as dry basis and fixed carbon (FC) was calculated by difference. VM includes those compounds which were driven-off as volatiles during heating while FC refers to the remaining constituents after the releases of volatiles. Moisture content of the feedstock had a marked effect on the conversion efficiency of pyrolysis processes, its energy content and the product yield. The elemental content of carbon, hydrogen and nitrogen in the algal biomass was 52.60%, 6.21% and 8.75% respectively obtained on dry basis. Oxygen content was calculated by difference. The H/C and O/C molar ratios were calculated from elemental composition as 1.42 and 0.46 respectively. A higher value of H/C molar ratio than O/C molar ratio indicated presence of high content of aliphatic hydrocarbon compounds and a small amount of polar compounds in *S. dimorphus*. The nitrogen content of the *S. dimorphus* was significantly higher compared to lignocellulosic biomass samples, most likely due to presence of protein content in sample [17].

3.1.2. Thermo-gravimetric analysis

TGA-DTG analysis of microalgae sample was carried out at a

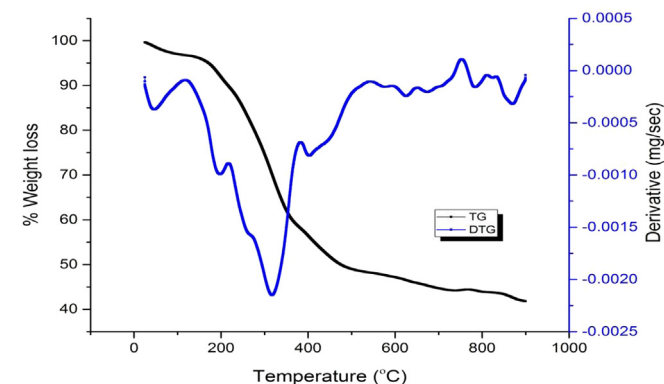


Fig. 1. Thermogravimetric (TG) and derivative thermogravimetric (DTG) curves of *S. dimorphus* at 10 °C/min heating rate.

temperature range of 50–900 °C with heating rate of 10 °C/min under nitrogen atmosphere and was shown in Fig. 1. TGA and DTG curves revealed the three stages of decomposition of the algae in the pyrolysis process; similar results were reported by Agarwal and Chakraborty [34]. Stage 1 was from temperature of 50 °C to the temperature of initial volatilization (180 °C), stage 2 was from 180 °C to the final devolatilization temperature (500 °C) and the third stage was from 500 to 900 °C. At the temperature of 320 °C the rate of reaction was maximum. In stage 1, the slight weight loss was due to the elimination of moisture and light volatile materials from the feedstock which was in conformity with Peng et al. [12] and Kumar et al. [33]. Based on the decomposition of biomass components, Stage 2 can be further divided into two zones. The first zone with a strong peak denoted the decomposition of proteins and carbohydrates while the second zone with the weak peak denoted the decomposition of lipids [12]. In the third stage, there was decomposition of carbonaceous product and thus, the loss in weight was mainly due to the process similar to gasification. In this zone highly non-volatile carbon compounds vaporized forming CO and CO₂ due to the high temperatures [34].

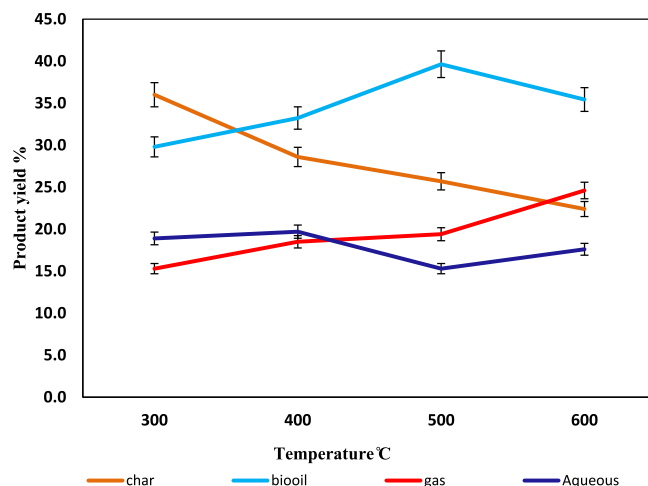


Fig. 2. Pyrolytic product yield of *S. dimorphus*.

3.2. Sample pyrolysis runs

The plot between the yield of bio-oil, biochar, gaseous and aqueous products vs. temperature has been shown in Fig. 2, obtained by pyrolysis of *S. dimorphus* at temperatures ranging from 300 to 600 °C with a heating rate of 40 °C/min under nitrogen atmosphere. The maximum bio-oil yield (39.6%) was obtained at the temperature of 500 °C, the yield however decreased to 35.4% at 600 °C. The temperature that yielded the highest bio-oil was considered as the optimum condition. The effect of temperature on product yields was also studied by many researchers [35–37] and they also reported the temperature of maximum liquid product yield as the optimum temperature/condition. The yield of synthetic fuel from tea waste was reported by Uzun et al. [35], who reported the maximum yield of 29.62 wt% at a temperature of 500 °C. Similarly, Shadangi and Mohanty [36] reported that the temperature of 550 °C which produced the maximum yield (55.17 wt%) of liquid product from karanja seed as the optimum condition. Majhi et al. [37], reported the maximum liquid product yield of 45 wt% at a temperature of 550 °C and this temperature was also reported as the optimum condition for pyrolysis of *J. curcas* cake.

The bio-oil obtained under optimum condition was further separated by adsorption chromatography. Weight percentage of all the sub-fractions obtained from microalgae bio-oils was found to be in equivalent amount. Yield of sub-fractions in wt% basis for the bio-oil were shown in Fig. 3. Bio-oil of the microalgae contained 18% of *n*-hexane fractions which could be useful for the production of hydrocarbons. Yields of toluene and ethyl acetate fractions of the bio-oil were found to be relatively more than that of the other two fractions (30% and 28%). After elution with ethyl acetate, methanol extraction of the bio-oil yielded polar compounds which were 24%.

The biochar yield reduced from 36.0% to 22.4% with increase in the pyrolysis temperature from 300 to 600 °C. The decrease in the char yield with increasing temperature could be either due to primary decomposition of the biomass at higher temperatures or through secondary decomposition of the char residue in the form of dehydrogenation and dehydration of reaction of hydroxyl groups. As the temperature increases from 300 to 600 °C the gaseous product yield also increased. The gas yield obtained was found to be maximum (24.6%) at 600 °C and minimum (15.3%) at 300 °C. The increase in gaseous products was thought to be predominantly due to secondary cracking of the pyrolysis vapors at higher temperatures. However, the secondary decomposition of the char at higher temperatures may also give non-condensable gaseous products, which could contribute to the increase in gas yield [38]. The

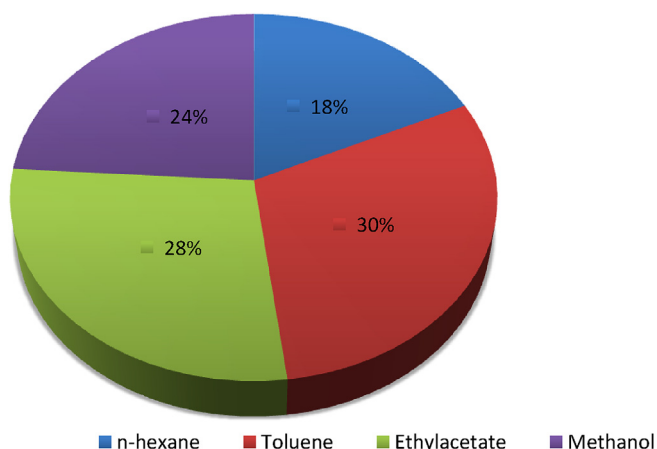


Fig. 3. Yields of column chromatographic fractions of *S. dimorphus* bio-oil.

aqueous phase obtained during pyrolysis was due to the inherent moisture in the biomass and dehydration reactions that took place during the pyrolysis process [39].

3.3. Characterization of the liquid product

3.3.1. Physical properties of liquid product

Table 2 shows the physical properties of the bio-oil obtained from the pyrolysis of *S. dimorphus* and a comparison with other transportation fuels. This liquid was dark brownish in color, dense and highly viscous which may affect pumping and injection of fuel in the engine at lower temperatures. Density and viscosity of bio-oil can be modified by blending it with commercial transportation fuels. The flash point of the bio-oil was similar with other fuels and a pour point of plus 3 °C will not cause any trouble in most of the regions except in colder regions with sub-zero climates, where it may lead to freezing problems. Calorific value of aliphatic sub-fraction of the bio-oil was found to be 33.42 MJ/kg which was higher than the calorific value of raw bio-oil (28.52 MJ/kg). This could be due to the elimination of oxygenated compounds from the bio-oil during *n*-hexane separation by column chromatography.

Elemental analysis of the bio-oil obtained from the pyrolysis of *S. dimorphus* showed that higher percentage of oxygen was present in the biomass than that of the bio-oil. The decrease in the oxygen content was favorable for production of transportation fuel as high oxygen content in transportation fuel disadvantageous. From the elemental analysis, the carbon, hydrogen and nitrogen contents of microalgae bio-oil obtained at 500 °C were found to be 74.73%, 10.60% and 5.79% respectively. The H/C molar ratio of the bio-oil was found to be 1.49, slightly higher than the original feedstock. The calorific value of the bio-oil was found to be 28.52 MJ/kg while the original biomass had calorific value of 19.0 MJ/kg. Similar findings were also reported for cherry seed pyrolysis where H/C ratios of bio-oil and raw material were found to be comparable while bio-oil had a comparatively higher calorific values than the raw material. The heating value of bio-oil depends mainly on oxygen content and the operating conditions of the pyrolysis process such as type of reactor, heating rate and reaction temperature [39].

3.3.2. Chemical composition of liquid product

FTIR spectra of the microalgae bio-oil sub-fractions were shown in Supplementary Fig. S1 which showed the presence of various functional groups. The absorption peak at 3200–3550 cm⁻¹ indicated the presence of oxygenated compounds along with moisture. This band was observed in methanol fractions of microalgae bio-oil. A sharp peak at 3580–3650 cm⁻¹ indicated the presence of free O–H group and this was observed in the ethyl acetate fractions of bio-oil. However, this band was not observed in the hexane and toluene fractions of bio-oil which indicated the absence of O–H group. This supports the fact that fractionation separates the oxygenated compounds from bio-oils and thus increases the aliphatic nature [40]. The C–H stretching vibrations between 2850 and 2925 cm⁻¹ and C–H deformation vibrations between 1410 and 1515 cm⁻¹ indicated the presence of alkanes. The C–H deformation vibrations were observed at 1378–1464 cm⁻¹ due to alkane groups. The presence of amines was detected by C–N stretching vibrations between 1080 and 1335 cm⁻¹. The C=O stretching vibrations between 1650 and 1750 cm⁻¹ was compatible with the presence of ketone, quinine and aldehyde groups etc. The absorbance of peaks between 1575 and 1675 cm⁻¹ representing C=C stretching vibrations is indicative of alkenes and aromatics. The region between 1210 and 1320 cm⁻¹ in methanol sub-fractions of bio-oil showed medium C–O stretching vibrations for carboxylic acids and their derivatives. The region between 700 and 900 cm⁻¹ contains various bands related to the aromatic, out of plane C–H bending. The high

Table 2Physico-chemical properties of *S. dimorphus* bio-oil and its comparison with petro-fuels.

Properties	Method	Bio-oil	Gasoline [31]	Diesel [31]
Appearance		Dark Brownish		
Density, 15 °C (g/cm ³)	ASTM D1298-99	0.8418		
Specific gravity at 15°C/15 °C	ASTM D1298-99	0.8425	0.72–0.78	0.82–0.85
p ^H		5	7	3.6–5.6
Pour point (°C)	ASTM D5853-09	+3	–40	–40 to –1
Cloud point (°C)	ASTM D1310-01(2007)	10		
Flash point (°C)	ASTM D6450-05(2010)	40	–43	53–80
Fire point (°C)	ASTM D1310-01(2007)	57	–	–
C	–	74.73	–	–
H	–	10.60	–	–
N	–	5.79	–	–
S	–	0.61	–	–
O	–	8.27	–	–
H/C	–	1.49	–	–
O/C	–	0.08	–	–
Calorific value (MJ/kg)	–	28.52	42–46	42–45

Table 3Percentage of hydrogen based on ¹H NMR analysis of *S. dimorphus* bio-oil sub-fractions.

Proton assignment	Chemical shift (ppm)	<i>n</i> -Hexane	Toluene	Ethyl acetate	Methanol
Aromatics	6.5–9.0	3.25	15.43	12.54	12.98
Phenolic OH or olefinic proton	5.0–6.5	3.90	5.43	6.88	3.89
Ring-join methylene (ArMACROBUTTON InsGlyph eCH ₂ –Ar)	3.3–4.5	0.45	1.17	3.83	8.90
CH ₃ , CH ₂ , CH α to an aromatic ring	2.0–3.3	11.89	11.79	16.50	29.27
CH ₂ , CH ₃ to an aromatic ring (naphthenic)	1.6–2.0	6.75	6.90	7.00	6.94
β -CH ₃ , CH ₂ , and CH ₂ or further from an aromatic ring	1.0–1.6	60.82	46.69	39.55	20.95
CH ₃ or further from an aromatic ring	0.5–1.0	10.67	8.86	9.71	6.05

intensity of the above mentioned bands in toluene sub-fraction indicated that the aromatic hydrogen was located in aromatic rings with high degree of substitution.

¹H NMR spectra were applied to the bio-oil fractions, *n*-hexane, toluene, ethyl acetate and methanol and their results were shown in [Supplementary Fig. S2](#). The hydrogen distributions of ¹H NMR on percentage basis were listed in [Table 3](#). From the NMR spectra, the most up-field region from 0.5 to 1.6 ppm, represented aliphatic protons that were attached to carbon atoms, at least two bonds, removed from a C=C double bond or heteroatom (O or N). This region was shown to be more populated for *n*-hexane fraction of microalgae bio-oil (72%) than other fractions. This indicated the presence of higher aliphatic content in *n*-hexane fractions of the bio-oil. The aliphatic proton percentages in toluene, ethyl acetate and methanol were lower than hexane fractions.

The integrated region from 1.6 to 3.3 ppm represented the protons on aliphatic carbon atoms that may be bonded to a C=C double bond (aromatic or olefinic) or are two bonds away from a heteroatom. Lower amount of proton percentage was observed in this region for hexane and toluene fractions than that of other fractions. Methanol fraction showed highest percentage of proton for this integral region.

¹H NMR spectrum from 3.3 to 4.5 ppm region represented protons on carbon atoms next to an aliphatic alcohol or ether, or a methylene group that joined two aromatic rings. Highest percentage of protons was observed in methanol fraction of bio-oil.

The region between 5.0 and 6.5 ppm represented the presence of phenolic OH or olefinic protons. In this region lowest proton percentage was observed for aliphatic fraction (*n*-hexane fraction) while it was the highest for ethyl-acetate fraction of bio-oil. This may be due to the fact that fractionation with non-polar solvent separated the oxygenated compounds from the bio-oil. Phenolic derivatives can be used as flavourings agents in the food industry.

The region from 6.5 to 9.0 ppm was assigned to aromatic

protons. This represented not only those hydrogen atoms in benzenoids, but also those in heteroaromatics containing O and N. For *n*-hexane fraction, the proton percentage (3%) in this region was quite lower than other fractions whereas the toluene contained the highest percentage of protons (15%). The ethyl acetate and methanol fractions showed almost equivalent amounts of proton percentage in this region. These solvents eluted mostly aromatic compounds including monocyclic and polycyclic aromatic hydrocarbons, phenolic and neutral oxygenated compounds respectively. Finally, methanol fraction extracted the polar molecules from the bio-oil. The region from 7.0 to 8.0 ppm was assigned for the amide molecules. Amines, esters, aldehyde, cyanide and nitro groups substituted molecules were observed in the region of 0.5–4.5 ppm. It was obvious from ¹H NMR spectra that aliphatic compounds present in the *n*-hexane fractions of bio-oil were relatively higher than other fractions. The absence of oxygen content in this fraction was also observed in the FTIR spectra. This was also supported by the higher calorific value (33.42 MJ/kg) of *n*-hexane fraction. Thus, fractionation reduced the disadvantages associated with the bio-oil that were highly oxygenated.

GC–MS analysis was carried out to identify the organic compounds present in the un-fractionated bio-oil and their corresponding fractions. TIC of the microalgae bio-oil and its sub-fractions were shown in [Supplementary Figs \(S3, S4, S5, S6 and S7\)](#). The compounds present in the bio-oils were identified by comparing the chromatograms of pyrolytic oils and their corresponding fractions with standard chromatographic data available from NIST 98 spectrum library. Library search showed that the bio-oil was a mixture of complex compounds therefore, a perfect separation of all the compounds was difficult. Fractionation of bio-oil with solvents of different polarity was able to separate some of the compounds which were listed in [Supplementary Table 1](#) along with the retention time, compound name and area percentage. Most of the compounds present in bio-oil were aromatic and

phenolic in nature. The presence of phenols was also evident from FTIR results. Aromatic hydrocarbons, heterocyclic, amide, indole, alkane and nitrile were found in the bio-oil. Presence of acids in bio-oil is undesirable due to its corrosive nature. Carbonyl groups are also oxygenated compounds and they affect the stability and heating value of bio-oil. Therefore, these compounds must be detached from bio-oil structure through efficient separation techniques so as to utilize the bio-oil effectively. The *n*-hexane fraction of bio-oil showed alkane as dominant group. Three types of compounds were identified in *n*-hexane fraction: *n*-alkanes, alkenes and branched hydrocarbons (isoprenoids). The majority of straight-chain alkanes were distributed in the range from C₉ to C₂₂; however, some part of the light hydrocarbons may have escaped through evaporation along with the solvent after fractionation. Though alkenes were present in *n*-hexane fractions, *n*-alkanes were dominant. Similarly, toluene fraction contained most of the aromatic compounds, ethyl-acetate fraction contained mostly phenolic group of compounds and methanol fraction showed the presence of polar compounds. These results are in consistent with the findings of Horne and Williams [38].

The properties of bio-oil produced from microalgae depend on its biochemical composition viz. lipids, proteins and carbohydrates. The major constituents of bio-oil comprise of phenol and its alkylated derivatives, heterocyclic N-containing compounds, long-chain fatty acids, alkanes and alkenes. Among these compounds, hydrocarbons are valuable components of bio-oil from fuel application point of view. Specifically, aromatic hydrocarbons serve as important industrial chemicals and transportation fuel additives to increase octane number. Proteins produce large amounts of N-containing heterocycles, pyrroles and indoles, carbohydrates produce cyclic ketones as well as phenols while lipids in microalgae being cracked into hydrocarbons during pyrolysis as well as fatty acids [41].

3.4. Characterization of the solid product

The proximate and ultimate analysis of microalgae biochar obtained at different temperatures with heating rate 40 °C/min were shown in Table 1. Unlike lignocellulosic biochar, algal biochar may contain lesser oxygen and carbon than the raw biomass. With the increase in temperature, H/C and O/C ratios of biochar decreased which was probably due to dehydration, decarboxylation and decarbonylation reactions. The oxygen containing functionalities are decomposed by decarbonylation and decarboxylation reactions followed by increase in proportion of aryl C and carbonization of biochar [42].

With increase in temperature, the pH and EC values of biochar increased which may be due to the separation of alkali salts from organic materials. Biochar may be used as a liming agent as it is capable to resolve the acidity problem of soil. The electrical conductivity (EC) of biochar obtained from pyrolysis of *S. dimorphus* was found to increase with the increase in temperature. Biochar with higher EC when applied to soil may harm plant growth by reducing water uptake and cause nutrient imbalance due to increased salinity [27]. Therefore, EC of biochar is an important parameter which must be taken into consideration prior to biochar application in soil as a soil amendment. Though biochar with high pH may be desirable for recovery of highly acidic soils, but, may also harm the soil due to its high EC. Therefore, the final pyrolysis temperature is important for favorable values of pH and EC of biochar for its use as a soil amendment. After pyrolysis the minerals present in original biomass were left as ash in the biochar. The ash content in biochar increased with increase in pyrolysis temperature, which may be due to the increase in relative concentration of Ca, Mg, K, P, S and Zn in ash content at higher temperature [42]. The

Table 4

Comparison of BET surface area for different feedstocks.

Sample	BET (m ² /g)	Reference
<i>S. dimorphus</i> biochar at 300 °C	1.72	—
<i>S. dimorphus</i> biochar at 400 °C	3.34	—
<i>S. dimorphus</i> biochar at 500 °C	123	—
<i>S. dimorphus</i> biochar at 600 °C	89	—
Corn stover biochar at 500 °C	3.1	Mullen et al. [42]
Jute dust biochar at 700 °C	4.27	Choudhury et al. [43]
<i>U. flexuosa</i> algae biochar at 450 °C	4.26	Bird et al. [44]
Kusum seed biochar 550 °C	8.12	Koul et al. [48]
Mistcanthus biochar at 400 °C	1.65	Kwapinski et al. [49]

major difference among the biochar obtained at different pyrolysis temperatures in this study were in their ash content, which is comparatively higher in amount than the biochar. This may be due to the presence of exogenous material (sand) that could not be removed from the raw algal biomass [44]. Further, surface morphology of the biochar was studied by Scanning Electron Microscopy (SEM). It is used for high magnification imaging of biochar obtained at 500 °C with a heating rate of 40 °C/min which was shown in Supplementary Fig. S8. SEM is a very widely used technique which confirmed the heterogeneous distribution of pores and presence of rough texture.

Surface area of biochar obtained at different temperatures was shown in Table 4. As the production temperature of biochar increased from 300 to 500 °C, the surface area of biochar also increased from 1.72 to 123 m²/g, however, at higher temperature (600 °C) the surface area of biochar was found to be reduced (89 m²/g). Biochar produced at higher temperature exhibited higher BET surface area. When the pyrolysis temperature was increased to 600 °C, the BET surface area reduced again, which might be due to restructuring taking place within the biochar or due to the commencement of ash melting at higher temperatures [45].

FTIR spectra of biochar samples (Supplementary Fig. S9) showed the changes of the functional groups on the surfaces of the biochar produced at different temperature (300, 400, 500 and 600 °C). As the pyrolysis temperature increased to 600 °C, the peaks tend to disappear showing a loss of carboxylate (—COO) group from biochar with increased temperature. The bands corresponding to O—H stretches (at wave numbers of 3358–3403 cm^{−1}, broad), aromatic rings 1657, aromatic out of plane C—H bending (various bands between 900 and 700 cm^{−1}) and C=O (1550–1653 cm^{−1}) were identified. Weak absorption bands at 2950 and 2850 cm^{−1} were observed and assigned to aliphatic C—H stretching which decreased with increase in temperature. The spectrum of the char obtained at 600 °C showed the disappearance of most of the bands and suggested that the char was mainly an aromatic polymer carbon atom. The absorption peaks at 780 and 1400 cm^{−1} in the spectra of biochar produced at 300 °C were assigned to the carboxylate (—COO) group [46].

One the most crucial parameter of biochar is carbon sequestration potential (CSP). Higher CSP means higher resistance ability of biochar against both biotic and abiotic degradation. The recalcitrance of biochar increased from 0.62 to 0.76 with increase in pyrolysis temperature, this increase might be attributed to increase in aromatic C and decrease in N [47]. Thus, biochar produced at lower temperature had lower recalcitrance which indicated that it can be easily mineralized by microorganism compared to biochar produced at higher temperature.

4. Conclusion

The pyrolysis of microalgae *S. dimorphus* was performed at the

temperature range of 300–600 °C in the fixed-bed reactor with heating rate of 40 °C/min. Maximum yield of the bio-oil was obtained at 500 °C with calorific value of 28.52 MJ/kg. Calorific value of *n*-hexane fraction of bio-oil was higher (33.42 MJ/kg) than the respective unfractionated bio-oil which showed the removal of oxygenated compounds. Thus, fractionation of bio-oil may be useful in bio-oil upgradation and recovery of value added chemicals. SEM micrograph revealed heterogeneous porous structure of biochar. Biochar can be used as soil amendment in correcting the soil acidity problem and thereby reducing the liming requirement and improving the general soil condition for sustainable soil management. It was also observed that biochar obtained at higher temperature was highly recalcitrant and could be used for C-sequestration and GHG mitigation. With high availability of bio-resources like *S. dimorphus* and other alike species, their convertibility to bio-oil and biochar through pyrolysis and multiple usability of the pyrolysis products, establishment of decentralized bio-refineries at village clusters in north-east India could be a viable option for production of bio-oil and biochar in a sustainable and environment friendly manner.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.renene.2016.03.081>.

References

- [1] U. Morali, S. Sensöz, Pyrolysis of hornbeam shell (*Carpinus betulus*L.) in a fixed bed reactor: characterization of bio-oil and bio-char, *Fuel* 150 (2015) 672–678.
- [2] L. Brennan, P. Owende, Biofuels from microalgae – a review of technologies for production, processing, and extractions of biofuels and co-products, *Renew. Sustain. Energy Rev.* 14 (2010) 557–577.
- [3] Q. Hu, M. Sommerfeld, E. Jarvis, M. Ghirardi, M. Posewitz, M. Seibert, A. Darzins, Microalgal triacylglycerols as feedstocks for biofuel production: perspectives and advances, *Plant J.* 54 (4) (2008) 621–639.
- [4] Y. Chisti, Biodiesel from microalgae, *Biotechnol. Adv.* 25 (3) (2007) 294–306.
- [5] S. Sudjito, N. Hamidi, U. Yanuher, I.N.G. Wardana, Potential and properties of marine microalgae *Nannochloropsis oculata* as biomass fuel feedstock, *Int. J. Energy Environ. Eng.* 5 (2014) 279–290.
- [6] R.S. Chutia, R. Katak, T. Bhaskar, Characterization of liquid and solid product from pyrolysis of *Pongamia glabra* deoiled cake, *Bioresour. Technol.* 165 (2014) 336–342.
- [7] K.Y. Chan, Z.L. Van, I. Meszaros, A. Downie, S. Joseph, Agronomic values of greenwaste biochar as a soil amendment, *Soil Res.* 45 (2007) 629–634.
- [8] J. Lehmann, Bio-energy in the black, *Front. Ecol. Environ.* 5 (2007) 381–387.
- [9] S. Jeffery, F.G.A. Verheijen, M. Velde, A.C. Bastos, A quantitative review of the effects of biochar application to soils on crop productivity using meta-analysis, *Agriculture, Ecosyst. Environ.* 144 (2011) 175–187.
- [10] Q.Y. Wu, B. Zhang, N.G. Grant, High yield of hydrocarbon gases resulting from pyrolysis of yellow heterotrophic and bacterially degraded *Chlorella protothecoides*, *J. Appl. Phycol.* 8 (1996) 181–184.
- [11] W.M. Peng, Q.Y. Wu, P.G. Tu, Effects of temperature and holding time on production of renewable fuel from pyrolysis of *Chlorella protothecoides*, *J. Appl. Phycol.* 12 (2000) 147–152.
- [12] W. Peng, Q. Wu, P. Tu, N. Zhao, Pyrolytic characteristics of microalgae as renewable energy source determined by thermogravimetric analysis, *Bioresour. Technol.* 80 (2001) 1–7.
- [13] X.L. Miao, Q.Y. Wu, C.Y. Yang, Fast pyrolysis of microalgae to produce renewable fuels, *J. Anal. Appl. Pyrolysis* 71 (2004) 855–863.
- [14] S. Grierson, V. Strezov, G. Ellem, R. McGregor, J. Herbertson, Thermal characterisation of microalgae under slow pyrolysis conditions, *J. Anal. Appl. Pyrolysis* 85 (2009) 118–123.
- [15] Z. Shuping, W. Yulong, Y. Mingde, L. Chun, T. Junmao, Pyrolysis characteristics and kinetics of the marine microalgae *Dunaliella tertiolecta* using thermogravimetric analyser, *Bioresour. Technol.* 101 (2010) 359–365.
- [16] X. Miao, Q. Wu, C. Yang, Fast pyrolysis of microalgae to produce renewable fuels, *J. Anal. Appl. Pyrolysis* 71 (2004) 855–863.
- [17] K. Chaiwong, T. Kiatsiriroat, N. Vorayos, C. Thararax, Study of bio-oil and bio-char production from algae by slow pyrolysis, *Biomass Bioenergy* 56 (2013) 600–606.
- [18] X. Miao, Q. Wu, High yield bio-oil production from fast pyrolysis by metabolic controlling of *Chlorella protothecoides*, *J. Biotechnol.* 110 (2004) 85–93.
- [19] A.F. Ferreira, A.P.S. Dias, C.M. Silva, M. Costa, Bio-oil and bio-char characterization from microalgal biomass, *MEFTE* (2014) 11–12.
- [20] M.M. Phukan, R.S. Chutia, B.K. Konwar, R. Katak, Microalgae *Chlorella* as a potential bio-energy feedstock, *Appl. Energy* 88 (10) (2011) 3307–3312.
- [21] R.C.D. Goswami, M.C. Kalita, Microalgal resources in Chandrapur area, North-East, Assam, India: a perspective for industrial refinement system and a boon for alternative energy generation and mitigation of green house gases, *Arch. Appl. Sci. Res.* 4 (2) (2012) 795–799.
- [22] H. Okkoy, M. Naddaf, A. Alinizam, M.F. Azmeah, Growth promotion of indigenous *Scenedesmus dimorphus* strain under different conditions using stirred tank photobioreactor, *Int. J. ChemTech Res.* 8 (11) (2015) 221–228.
- [23] A. Cici, M. Bravi, Production of the freshwater microalgae *Scenedesmus dimorphus* and *arthrosira platensis* by using cattle digestate, *Chem. Eng. Trans.* 38 (2014) 85–90.
- [24] A. Pons, P. Roca, C. Aguilo, F.J. Garcia, M. Alemany, A. Paloo, A method for the simultaneous determination of total carbohydrate and glycerol in biological samples with the anthrone reagent, *J. Biochem. Biophys. Methods* 4 (1980) 227–231.
- [25] O.H. Lowry, N.J. Rosebrough, A.L. Farr, R.J. Randall, Protein measurement with the Folin phenol reagent, *J. Biol. Chem.* 193 (1951) 265–275.
- [26] E.G. Bligh, W.J. Dyer, A rapid method for total lipid extraction and purification, *Can. J. Biochem. Physiol.* 37 (1959) 911–917.
- [27] N. Bordoloi, R. Narzari, R.S. Chutia, T. Bhaskar, R. Katak, Pyrolysis of *Mesua ferrea* and *Pongamia glabra* seed: characterization of bio-oil and its sub-fractions, *Bioresour. Technol.* 178 (2015) 83–89.
- [28] S. Naik, V.V. Goud, P.K. Rout, A.K. Dalai, Supercritical CO₂ fractionation of bio-oil produced from wheat–hemlock biomass, *Bioresour. Technol.* 101 (2010) 7605–7613.
- [29] O.R. Harvey, L.J. Kuo, A.R. Zimmerman, P. Louchouart, J.E. Amonette, B.E. Herbert, An index-based approach to assessing recalcitrance and soil carbon sequestration potential of engineered black carbons (Biochars), *Environ. Sci. Technol.* 46 (2012) 1415–1421.
- [30] L. Zhao, X. Cao, O.R.M. Sek, A. Zimmerman, Heterogeneity of biochar properties as a function of feedstock sources and production temperatures, *J. Hazard. Mater.* 256–257 (2013) 1–9.
- [31] N.K. Nayan, S. Kumar, R.K. Singh, Characterization of the liquid product obtained by pyrolysis of karanja seed, *Bioresour. Technol.* 124 (2012) 186–189.
- [32] V.K. Singh, A.B. Soni, S. Kumar, R.K. Singh, Pyrolysis of saal seed to liquid product, *Bioresour. Technol.* 151 (2014) 432–435.
- [33] R.R. Kumar, D. Ramesh, T. Mutanda, I. Rawat, F. Bux, Thermal behavior and pyrolytic characteristics of freshwater *Scenedesmus* sp., *Biomass. Energy Sources Part A* 37 (2015) 1383–1391.
- [34] A. Agrawal, S. Chakraborty, A kinetic study of pyrolysis and combustion of microalgae *Chlorella vulgaris* using thermo-gravimetric analysis, *Bioresour. Technol.* 128 (2013) 72–80.
- [35] B.B. Uzun, E.A. Varol, F. Ates, N. Özbay, E.A. Pütün, Synthetic fuel production from tea waste: characterisation of bio-oil and bio-char, *Fuel* 89 (2010) 176–184.
- [36] K.P. Shadangi, K. Mohanty, Thermal and catalytic pyrolysis of Karanja seed to produce liquid fuel, *Fuel* 115 (2014) 434–442.
- [37] A. Majhi, Y.K. Sharma, D.V. Naik, R. Chauhan, The production and evaluation of bio-oil obtained from the *Jatropha curcas* cake, *Energy Sources Part A Recovery, Util. Environ. Eff.* 37 (2015) 1782–1789.
- [38] P.A. Horne, P.T. Williams, Influence of temperature on the products from the flash pyrolysis of biomass, *Fuel* 75 (9) (1996) 1051–1059.
- [39] G. Duman, C. Okutucu, S. Ucar, R. Stahl, J. Yanik, The slow and fast pyrolysis of cherry seed, *Bioresour. Technol.* 102 (2011) 1869–1878.
- [40] R. Saikia, R.S. Chutia, R. Katak, K.K. Pant, Perennial grass (*Arundo donax* L.) as a feedstock for thermo-chemical conversion to energy and materials, *Bioresour. Technol.* 188 (2015) 265–272.
- [41] W.H. Chena, B.J. Lin, M.Y. Huang, J.S. Chang, Thermochemical conversion of microalgal biomass into biofuels: a review, *Bioresour. Technol.* 184 (2015) 314–327.
- [42] C.A. Mullen, A.A. Boateng, N.M. Goldberg, I.M. Lima, D.A. Laird, K.B. Hicks, Bio-oil and bio-char production from corn cobs and stover by fast pyrolysis, *Biomass Bioenergy* 34 (2010) 67–74.
- [43] N.D. Choudhury, R.S. Chutia, T. Bhaskar, R. Katak, Pyrolysis of jute dust: effect of reaction parameters and analysis of products, *J. Mater. Cycles Waste Manag.* 16 (3) (2014) 449–459.
- [44] M.I. Bird, C.M. Wurster, P.P. Silva, A.M. Bass, R. Nys, Algal biochar – production and properties, *Bioresour. Technol.* 102 (2011) 1886–1891.
- [45] F. Ronsse, S.V. Hecke, D. Dickinson, W. Prins, Production and characterization of slow pyrolysis biochar: influence of feedstock type and pyrolysis conditions, *Glob. Change Biol. Bioenergy* 5 (2013) 104–115.
- [46] A.E. Pütün, N. Özbay, E.P. Önal, E. Pütün, Fixed-bed pyrolysis of cotton stalks for liquid and solid products, *Fuel Process. Technol.* 86 (11) (2005a) 1207–1219.
- [47] K.A. Spokas, W.C. Koskinena, J.M. Bakera, D.C. Reicosky, Impacts of woodchip

- biochar additions on greenhouse gas production and sorption/degradation of two herbicides in a Minnesota soil, *Chemosphere* 77 (2009) 574–581.
- [48] M. Koul, K.P. Shadangi, K. Mohanty, Thermo-chemical conversion of Kusum seed: a possible route to produce alternate fuel and chemicals, *J. Anal. Appl. Pyrolysis* 110 (2014) 291–296.
- [49] W. Kwapinski, C.M.P. Byrne, E. Kryachko, P. Wolfram, C. Adley, J.J. Leahy, E.H. Novotny, M.H.B. Hayes, Bio-char from biomass and waste, *Waste Biomass Valorization* 1 (2010) 177–189.